Lists and Real Estate Analyzer Using Files

Important AI Statement: Students are strictly prohibited from using AI tools (ChatGPT, Copilot, or any other code-generating software) to complete this assignment. All code must be written manually by the student. Any detected AI-generated code will result in a failing grade for that assignment.

- **Requirement:** Submit only your own original code. Do NOT copy, paste, or modify code generated by AI. You may refer to textbooks, class notes, or personal experimentation only.
- **Honor Statement:** By submitting this assignment, you affirm that all code is your own work and was not created using AI code generation tools.

Penalty Statement: Violations of this policy will be treated as academic dishonesty, subject to the institution's academic integrity rules and 0 for this assignment.

The owner of BDJ Real Estate of Springfield has hired you to design a sales analyzer program for real estate sales. You have been given a file called **RealEstateData.csv** that has this data:

- Street Location data type is string
- City data type is string
- Zip data type is string
- Bedroom count is an integer
- Bathroom count is an integer
- Square Footage is an integer
- Property Type is a string
- Price is a float
- Latitude is a string
- Longitude is a string

Once the data input is complete you need to write code to find out the different analytics for the sales: Min, Max, Average, Median, Total.

You will also have to get a property total by City and by Property Type. To accomplish this you will need some lists and dictionaries. How many you need of each is up to you ☺

1. This assignment is for learning purposes you cannot use AI/ChatGPT to create the code.

2. Put **s** prefixes in front of variables that contain string values and **i** prefix for variables that contain integer numbers and **f** for floating numbers. Add your initials such as BLC at the end of each variable (not constants). For example: fSalesPriceBLC.

3. In a function called **getDataInput** you will need to read the entire file in. In Blackboard there are several different ways to process a file. Seeing this is a .csv (comma separated values) you should know which file read sample to use to accomplish this.

4. Code a function called **getMedian** that receives a list and returns a float that does the following (you must write your own code you can **NOT** use the median function in the Python statistics module):
    - To find the median it is the value that is exactly in the middle. Therefore you should sort this list before doing anything else.
    - If the number of entries in the list is odd, divide the count by 2 and use that entry as the median
    - If the number of entries in the list is even divide the count by 2. Take that that entry and the entry before it and average the two elements and use that as the median.
    - Return the calculated median value.

5. Code a **main** function that includes these requirements:
    - Call the **getDataInput** function to get the data you need for this program.

- Code a loop that reads each records from the list you got in the previous step.
  - Using Python's string methods/functions extract these columns from the record:
    1. City
    2. Property Type
    3. Zip
    4. Price
  - Each **Price** that is extracted must be added to a list you will be using to get summary functions such as Min, Max, Avg and Median.
  - For each **City** total up the **Price**. Each city should only have one summary.
  - For each **Zip** total up the **Price**. Each Zip should only have one summary.
  - For each **Property Type** total up the **Price**. Each Type should only have one summary.

- From the Property Prices list determine the **Minimum** value and output formatted as currency with 2 decimal positions to the screen.
- From the Property Prices list determine the **Maximum** value and output formatted as currency with 2 decimal positions to the screen.
- From the Property Prices list determine the **Total** value and output formatted as currency with 2 decimal positions to the screen.
- From the Property Prices list calculate the **Average** and output currency with 2 decimal positions to the screen.
- From the Property Prices list the **Median** by calling the **getMedian** function you coded in Step 2 by passing list of values that was populated in the loop. Receive the Median value returned from the getMedian function and output formatted value as currency with 2 decimal positions.

6. Output ALL the Summaries by City, Zip and Property Types (see sample output below, but, not the Zip summary is not shown so you will have to wing it).

7. Use comments to document your code. At the top of your code place a Comment Heading Block that has these elements: Your Name, Code Name and you must answer these reflection questions:
   - Share what did you liked about this assignment?
   - Share what you struggled with?
   - Which approach did you use to import the data and why did you choose it?
   - How many dictionaries did you use for this assignment?
   - Share your top 2 things you learned on this assignment

```
# Name: YourName
# Assignment: Real Estate and Files
# Reflection:  Share what you liked about this assignment?
#              Share what you struggled with?
#              Which approach did you use to import the data and why did you choose it?
#              How many dictionaries did you use for this assignment?
#              Share exactly 2 things you learned on this assignment:
#               1.
#               2.
```

Sample Output

```
Minimum                   1,551.00
Maximum                 884,790.00
Sum                 230,632,100.00
Avg                     234,144.26
Median                  213,750.00
```

Summary by Property Type

```
Summary by Property Type:
Residential          219,333,711.00
Condo                  8,104,438.00
Multi-Family           2,918,951.00
Unkown                   275,000.00
```

Summary by City

```
Sacramento           86,806,099.00
Rancho Cordova        7,375,366.00
Rio Linda             2,245,459.00
Citrus Heights        6,549,022.00
North Highlands       2,848,846.00
Antelope              7,672,381.00
Elk Grove            30,911,977.00
Elverta                 531,464.00
Galt                  4,975,812.00
Carmichael            5,913,695.00
Orangevale            3,070,755.00
Folsom                7,054,323.00
Mather                  237,800.00
Pollock Pines           720,908.00
Gold River            1,432,000.00
El Dorado Hills      11,309,076.00
Rancho Murieta          893,250.00
Wilton                3,087,542.00
Greenwood               395,000.00
Fair Oaks             2,731,506.00
Cameron Park          2,411,500.00
Lincoln               6,950,827.00
Placerville           3,638,634.00
Meadow Vista            230,000.00
Roseville            15,577,356.00
Rocklin               6,491,209.00
Auburn                2,029,454.00
Loomis                1,134,000.00
El Dorado               494,000.00
Penryn                  506,688.00
Granite Bay           2,036,200.00
Foresthill              194,818.00
Diamond Springs         216,033.00
Shingle Springs         275,000.00
Cool                    300,000.00
Walnut Grove            380,000.00
Garden Valley           490,000.00
Sloughhouse               2,000.00
West Sacramento         512,100.00
```

Grading Rubric

Criteria	Meets (100%)	Somewhat (50%)	Not Present (0%)
Data is imported from the file 20 points	Data Import Function fully coded.	Data Import Function somewhat coded.	Data Import Function not present. Data validation not implemented.

	Data validation was fully implemented and functional.	Data validation was attempted but not fully functional.	
Coded the getMedian function as detailed in the instructions 10 points	The getMedian Function is present and coded correctly.	The getMedian Function was attempted but with some errors or incorrect results or not coded in the most efficient manner in terms of execution.	The getMedian function is not present and/or the median logic was all placed in the main() function.
Loop, List Coding and Appending Function 30 points	Loop, List and Appending were done properly with the correct results using Python list functions.	Loop, List and Appending were attempted but with some errors or incorrect results. Or Python list functions not properly used.	Loop, List and Appending were not attempted.
Summary Functions 20 points	Summary Functions are all present and coded correctly.	Summary Functions were attempted but with some errors or incorrect results or not coded in the most efficient manner in terms of execution.	Summary Functions not attempted.
Comment Block Reflections, Comments and Variable Prefixing and Suffixing 20 points	Comment Heading Block is present and reflection responses are present. Comments present and reflection questions are fully answered and variable prefixing and initial suffixing.	Either Comment Heading Block/Comments present but answered vaguely or copied or Variable prefixing/suffixing is present	No Comment Heading Block/Comments present and No variable prefixing/suffixing.